Step 2a - H7 - Ore weakening before comminution

Market scan

Market size

The relevant market is comminution and ore pre-treatment equipment within the broader mineral processing equipment market (~$23B/yr in 2025). Crushing, screening and grinding equipment alone is roughly $10-15B/yr worldwide, growing ~4-6%/yr, and comminution consumes ~36-50% of total mine-site energy, making it the single biggest energy consumer and the natural target for energy-saving retrofits. A microwave pre-treatment unit would sell as an add-on upstream of mills into this existing spend, plus a large opex-saving value pool: comminution electricity on a big copper concentrator is worth tens of $M/yr per site.

Region	Size ($B)	Region	Size ($B)
Europe	~$2-3B/yr (~10% of mineral processing equipment spend)	Asia	~$9-10B/yr (~40-45%; China, Australia, India dominant)
America	~$5-6B/yr (~25%; Chile/Peru copper belt + North America)	Africa	~$3-4B/yr incl. Middle East (~15%; fastest-growing, ~10% CAGR)

Target customers

ICMM's Global Mining Dataset lists ~15,000 mining/processing facilities across 151 countries; the addressable subset for ore weakening is hard-rock concentrators (copper, gold, iron ore, PGM, lithium) — order of ~3,000-5,000 sites. Economic value concentrates in a few hundred large concentrators (>10 Mt/y) run by majors and mid-tiers. Large miners are reachable via OEM/EPCM channels but are conservative, capex-disciplined and slow (multi-year pilot-before-adoption cycles); medium miners are the realistic early adopters; small mines lack capex and power infrastructure for high-MW microwave units.

Customer size	Approx. number	Market share (in %)
Large	~200-400 concentrators (>10 Mt/y ore)	~70-80% of world processed tonnage
Medium	~1,000-1,500 concentrators (1-10 Mt/y)	~15-25% of tonnage
Small	~3,000+ sites (<1 Mt/y)	<10% of tonnage

Players offering a solution today

Comminution equipment is supplied by a concentrated tier of global OEMs — Metso (#1 crushing, top-3 grinding), FLSmidth, Weir (Enduron HPGR), Sandvik, thyssenkrupp, CITIC — with the top 3 holding ~35% of crushing/screening and strong aftermarket lock-in. There is no established commercial supplier of microwave ore pre-treatment; the space is emerging (University of Nottingham + Teledyne e2v 'MicroHammer' with Rio Tinto at pilot scale; SELFRAG's high-voltage pulse machines are the closest commercialised analogue). Ore pre-treatment is therefore a contested white space the incumbents do not yet own.

Player	Type of solution	Market share (in %)

Large (Metso, FLSmidth, Weir, Sandvik, thyssenkrupp, CITIC)	Full comminution lines: gyratory/cone crushers, SAG/ball mills, HPGR, screens + aftermarket service; energy-efficient grinding packages	top 3 ~35% of crushing/screening; ~60-70% combined in large mills
Medium (Koppen, TAKRAF, Tenova, Multotec, Loesche, McLanahan)	HPGR, specialised mills, screening, regional OEM equipment and plant packages	~20-30%
Small/emerging (SELFRAG HVP, Teledyne e2v MicroHammer, Lightning Machines, ore-sorting vendors)	Novel pre-treatment/pre-concentration: high-voltage pulse fragmentation, microwave pilot systems, sensor sorting	<5%, no dominant player

Regulation

Regulation	Jurisdiction	What it means for us
EU Industrial Emissions Directive / Mining BREF (MIN BREF)	EU	Specific energy consumption per tonne of ore is being adopted as a key environmental indicator for mine sites; BAT conclusions will push operators toward best-available energy-efficient comminution techniques — favours retrofit pre-treatment.
EU ETS + Energy Efficiency Directive (Art. 11 audits)	EU	Carbon price (~EUR60-80/t CO2) and mandatory 4-yearly energy audits for large companies raise the cost of comminution electricity and force audited efficiency measures — strengthens the energy-saving sales case.
EU Critical Raw Materials Act (2024)	EU	10% of EU extraction and 40% of processing to be domestic by 2030 with fast-tracked permitting — creates new EU concentrator builds where a Swiss technology supplier can position early.
Australia Safeguard Mechanism	Australia	Large emitting facilities (incl. big mines) must reduce baseline emissions ~4.9%/yr to 2030; comminution electrification/efficiency is a primary lever on grid-powered sites.

EMF/RF safety: CISPR 11/EN 55011 ISM limits + EU Directive 2013/35/EU	EU/international	High-power microwave units must be shielded to ISM leakage limits at 915 MHz/2.45 GHz and worker EMF exposure limits — adds engineering/compliance requirements to machine design but no licensing barrier.
ICMM member net-zero 2050 commitments	Global (voluntary)	Major miners representing ~1/3 of the industry have committed to scope 1-2 net zero — pulls demand for step-change energy technologies beyond regulatory minimums.

Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input	Personal input
Name	Devin (agent draft)	Devin (agent draft)	Date	2026-10-30	2026-10-30	Name			Date	
Decision	Decision	Go to next step	Go to next step	Go to next step	Rework	Rework	Rework	Drop the hypothesis	Drop the hypothesis	Drop the hypothesis